

Revised Generation Interconnection System Impact Study Report

For Queue Project Q18-1061

Mcclellan Ar 138 kV
September 30, 2019

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q18-1061. The Interconnection Customer proposes a 89.4 MW wind facility interconnecting at the Mcclellan Ar 138 kV substation in the Kentucky area. The request was received on September 11, 2018. The study evaluated system intact and single contingency conditions using the regional planning case, including 427 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 25 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$48,000,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

For the loss of Wellington / Davisville 138.0 kV, the Davisville / Miabel 138.0 kV L loads to 193.5% of its 217 MVA rating. Project DFAX: 6.0%.

For the loss of Wellington / Davisville 138.0 kV, the Mirabel / Wec 138.0 kV line loads to 192.3% of its 222 MVA rating. Project DFAX: 6.0%.

The Cpu Silicon / Lanham 138.0 kV is overloaded to 159.1% of its 175 MVA emergency rating for the loss of the Lee County Solid Waste Energy / Cooper 345.0 kV. The project contributes 7.5% DFAX (6.7 MW) to this overload.

The Cpu Silicon / Kings Row 138.0 kV Line is overloaded to 146.6% of its 202 MVA emergency rating for the loss of the Lee County Solid Waste Energy / Cooper 345.0 kV. The project contributes 7.5% DFAX (6.7 MW) to this overload.

Spring Creek Rd / Hrris 138.0 kV: 129.3% loading (400 MVA) following the outage of the Massapequa / Wansley Combined Cycle 138.0 kV L; DFAX = 6.8%.

For the loss of Blackrock / Shawville Station 345.0 kV Line, the Carbon Center 500/345kV Transformer Bank loads to 125.8% of its 1,716 MVA rating. Project DFAX: 5.5%.

El Paso / Ravenna 500.0 kV L: 120.0% loading (2,948 MVA) following the outage of the Muskingum River / Tidd Substation 500.0 kV L; DFAX = 11.6%.

The Orchard Rd / Shawville Station 500.0 kV is overloaded to 119.8% of its 2,560 MVA emergency rating for the loss of the Hilltop Substation / Piney Creek Project 500.0 kV. The project contributes 7.4% DFAX (6.6 MW) to this overload.

Barrett / Norwich 500.0 kV: 119.4% loading (3,374 MVA) following the outage of the Beaver / Trego 500.0 kV; DFAX = 23.5%.

The Medora / Hampton Rd 345.0 kV Line is overloaded to 117.1% of its 1,928 MVA emergency rating for the loss of the Simpson Stuart Tap / Fishkill Plains 345.0 kV. The project contributes 6.8% DFAX (6.1 MW) to this overload.

The Hylebos / Ravenna 500.0 kV line is overloaded to 116.2% of its 2,504 MVA emergency rating for the loss of the Muskingum Riv / Tidd 500.0 kV. The project contributes 7.6% DFAX (6.8 MW) to this overload.

For the loss of Allegheny Energy Units 3 4 & 5 345 kV/1 kV Transformer, the Burwell 500 kV/345 kV Auto loads to 114.5% of its 1,191 MVA rating. Project DFAX: 5.6%.

Talenenergy Martins Crk Llc Harrisburg 500/345 kV Transformer Bank: 114.5% loading (1,584 MVA) following the outage of the Coronet / Westwood Generation Llc 500.0 kV L; DFAX = 5.2%.

Wolfe / Barrett 500.0 kV: 113.9% loading (3,201 MVA) following the outage of the Beaver / Trego Substation 500.0 kV line; DFAX = 17.4%.

The Hilltop 500/345kV Transformer Bank is overloaded to 113.6% of its 2,218 MVA emergency rating for the loss of the Talenenergy Martins Creek Llc Williamsport / Hilltop 500.0 kV. The project contributes 6.1% DFAX (5.5 MW) to this overload.

For the loss of Barrett / Norwich Substation 500.0 kV, the Minco / Barrett 230.0 kV loads to 112.9% of its 620 MVA rating. Project DFAX: 8.1%.

The Tule / Granite City Works 138.0 kV is overloaded to 112.2% of its 196 MVA emergency rating for the loss of the Davisville / N Appleton 138.0 kV. The project contributes 5.9% DFAX (5.3 MW) to this overload.

For the loss of Spatsum Lornex / Latah Junction 138.0 kV, the Frontier / Harvest Power Orlando 138.0 kV loads to 111.8% of its 304 MVA rating. Project DFAX: 50.1%.

The Larson St Substation / Norwich 230.0 kV is overloaded to 111.6% of its 1,633 MVA emergency rating for the loss of the Barrett / Norwich 500.0 kV. The project contributes 8.1% DFAX (7.3 MW) to this overload.

Frontier / Yuba City ta A 138.0 kV Line: 111.3% loading (405 MVA) following the outage of the Spatsum Lornex / Mcclellan Ar 138.0 kV; DFAX = 77.1%.

The Wharton / Toledo Bend 230.0 kV is overloaded to 111.3% of its 702 MVA emergency rating for the loss of the Omni-Hackberry / Norwich 500.0 kV L. The project contributes 8.5% DFAX (7.6 MW) to this overload.

The Ravenna 230/1kV TX is overloaded to 111.2% of its 1,411 MVA emergency rating for the loss of the Hylebos / Ravenna 500.0 kV L. The project contributes 5.8% DFAX (5.2 MW) to this overload.

For the loss of Mcclellan Ar / Spatsum Lornex 138.0 kV, the Westin-Elgin / Yuba City Sta A 138.0 kV loads to 110.7% of its 418 MVA rating. Project DFAX: 77.1%.

The 15Th Avenue South / Lucent 230.0 kV is overloaded to 110.3% of its 895 MVA emergency rating for the loss of the Omni-Hackberry / Norwich 500.0 kV. The project contributes 5.3% DFAX (4.8 MW) to this overload.

The Ravenna 500/1kV Autotransformer is overloaded to 109.8% of its 1,428 MVA emergency rating for the loss of the Hylebos / Ravenna 500.0 kV. The project contributes 5.8% DFAX (5.2 MW) to this overload.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Mirabel / Davisville 138.0 kV	Rebuild line with higher-capacity conductor	\$17,392,000	2% (\$344,000)
Mirabel / Wec 138.0 kV line	Rebuild line with higher-capacity conductor	\$8,493,000	2% (\$168,000)
Cpu Silicon / Lanham 138.0 kV	Rebuild line with higher-capacity conductor	\$3,021,000	4% (\$118,000)
Cpu Silicon / Kings Row 138.0 kV	Rebuild line with higher-capacity conductor	\$13,946,000	4% (\$546,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Spring Creek Rd / Hrris 138.0 kV	Rebuild line with higher-capacity conductor	\$4,320,000	2% (\$102,000)
Carbon Ctr 500-345 kV Transformer Bank	Replace transformer with higher-rated unit	\$32,811,000	2% (\$699,000)
El Paso / Ravenna 500.0 kV L	Rebuild line with higher-capacity conductor	\$200,693,000	2% (\$4,233,000)
Orchard Rd / Shawville Station 500.0 kV	Rebuild line with higher-capacity conductor	\$165,416,000	2% (\$3,527,000)
Barrett / Norwich 500.0 kV	Rebuild line with higher-capacity conductor	\$68,187,000	2% (\$1,428,000)
Medora / Hampton Rd 345.0 kV Line	Rebuild line with higher-capacity conductor	\$25,291,000	2% (\$546,000)
Hylebos / Ravenna 500.0 kV line	Rebuild line with higher-capacity conductor	\$112,027,000	2% (\$2,346,000)
Burwell 500-345 kV TX	Replace transformer with higher-rated unit	\$28,323,000	2% (\$610,000)
Talenenergy Martins Crk Llc Harrisburg 500-345 kV Transformer	Replace transformer with higher-rated unit	\$30,682,000	2% (\$653,000)
Wolfe / Barrett 500.0 kV	Rebuild line with higher-capacity conductor	\$137,654,000	2% (\$2,405,000)
Hilltop 500/345kV Transformer Bank	Replace transformer with higher-rated unit	\$19,757,000	2% (\$421,000)
Minco / Barrett 230.0 kV line	Rebuild line with higher-capacity conductor	\$10,975,000	2% (\$234,000)
Tule / Granite City Works 138.0 kV	Rebuild line with higher-capacity conductor	\$24,340,000	4% (\$896,000)
Frontier / Harvest Power Orlando 138.0 kV	Rebuild line with higher-capacity conductor	\$26,251,000	34% (\$8,939,000)
Norwich Substation / Larson St 230.0 kV	Rebuild line with higher-capacity conductor	\$12,404,000	2% (\$258,000)
Frontier / Yuba City ta A 138.0 kV Line	Rebuild line with higher-capacity conductor	\$17,455,000	45% (\$7,852,000)
Wharton / Toledo Bend 230.0 kV	Rebuild line with higher-capacity conductor	\$18,819,000	2% (\$435,000)
Ravenna 230/1kV TX	Replace transformer with higher-rated unit	\$8,308,000	3% (\$214,000)
Westin-Elgin Substation / Yuba City Sta A 138.0 kV	Rebuild line with higher-capacity conductor	\$21,154,000	45% (\$9,517,000)
15Th Avenue South / Lucent 230.0 kV	Rebuild line with higher-capacity conductor	\$21,591,000	3% (\$646,000)
Ravenna 500 kV/1 kV Autotransformer	Replace transformer with higher-rated unit	\$33,463,000	3% (\$863,000)
	Total allocated to project		\$48,000,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.